

InstantCoFFEA: Fast and accurate Tor flow correlation at realistic scale

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Motivation

1. $O(N^2)$ bottleneck problem
2. Not defined N (the scale of flows) and base rate

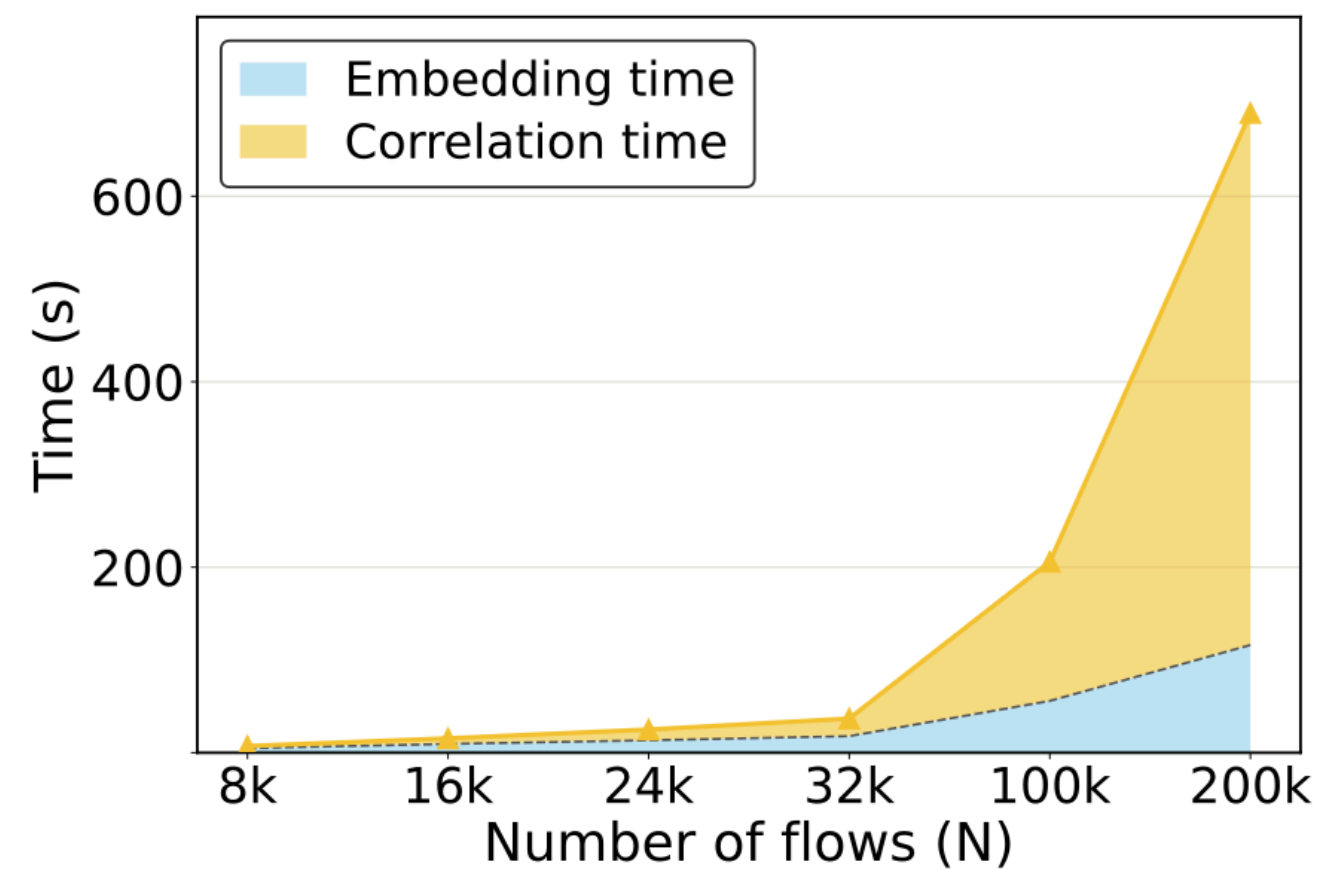
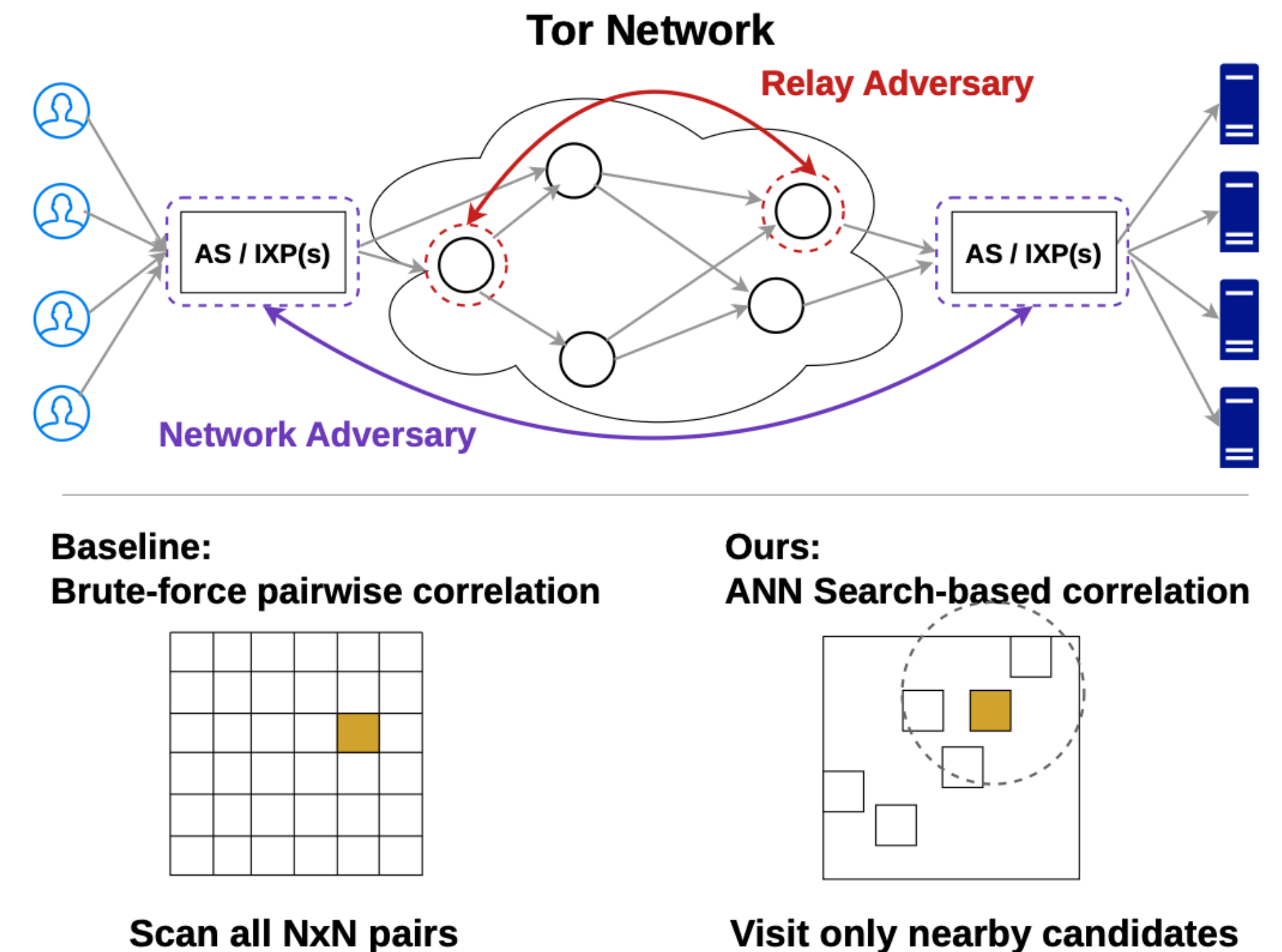


Fig. 1: Wall-clock time (seconds) of DeepCoFFEA as the number of flow pairs (N) increases.



Our approach

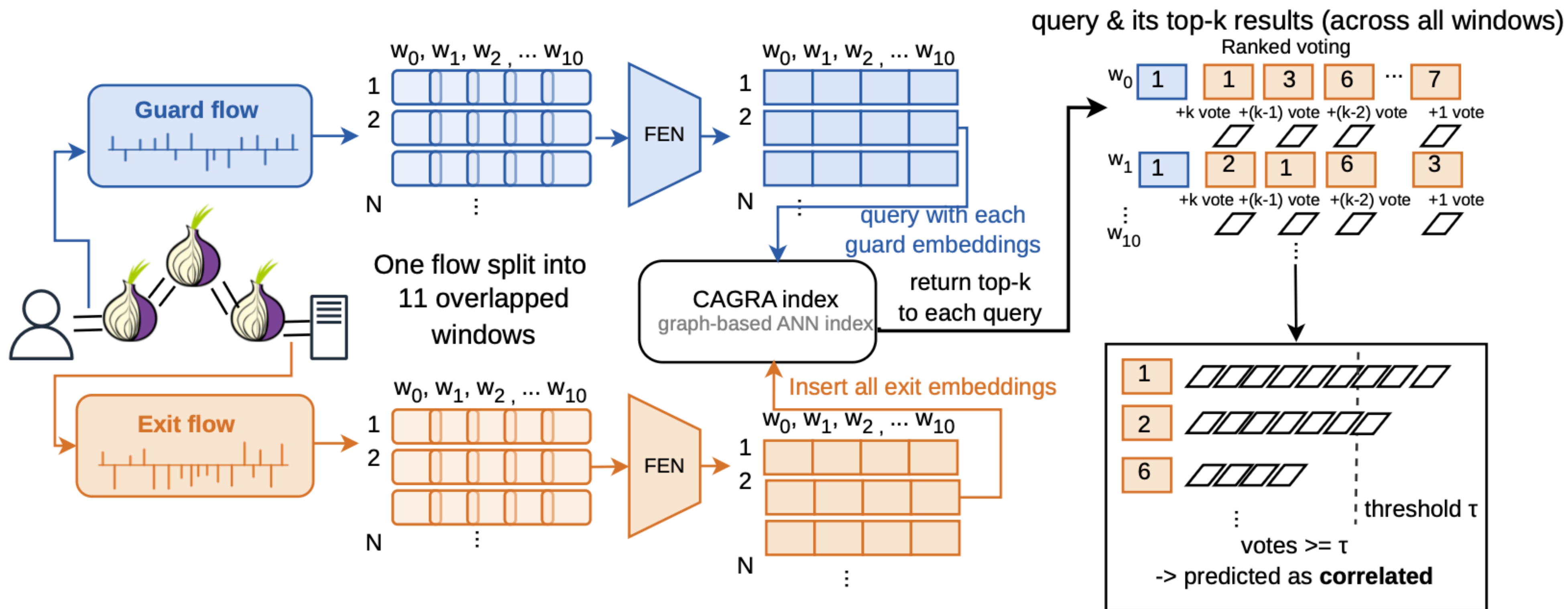
1. InstantCoFFEA:

1. To mitigate the $O(N^2)$ bottleneck, -> **integrated** with approximate nearest neighbor **(ANN) search** CAGRA.
2. Further, we improve the **voting scheme** -> achieve **better detection performance**

2. For realistic range of N and base rate

1. Given number of circuit in total network for 10 minute / Tor consensus document, we simulate the **path selection algorithm** to see how **each flow distribute over the network**

InstantCoFFEA



InstantCoFFEA

Theoretical Time Complexity Analysis

1. Time Complexity analysis per stage in pipeline
2. Operational cost comparison with hyperparameters

Table 1: Per-stage time complexity of DeepCoFFEA and InstantCoFFEA. Here, $W = 11$, and d, k, p , and $I_{\max}$ are treated as constants.

DeepCoFFEA		InstantCoFFEA	
Embedding	$O(WN)$	Embedding	$O(WN)$
Sim. Comp.	$O(WN^2)$	Index Build	$O(WNd^2 \log_d N + WNd^3 + WNd)$
Sorting	$O(WN^2 \log N)$	Searching	$O(WNI_{\max}pd)$
Voting	$O(N^2W + NWk \log Wk)$	Voting	$O(NWk)$
Total	$O(N^2 \log N)$	Total	$O(N \log N)$

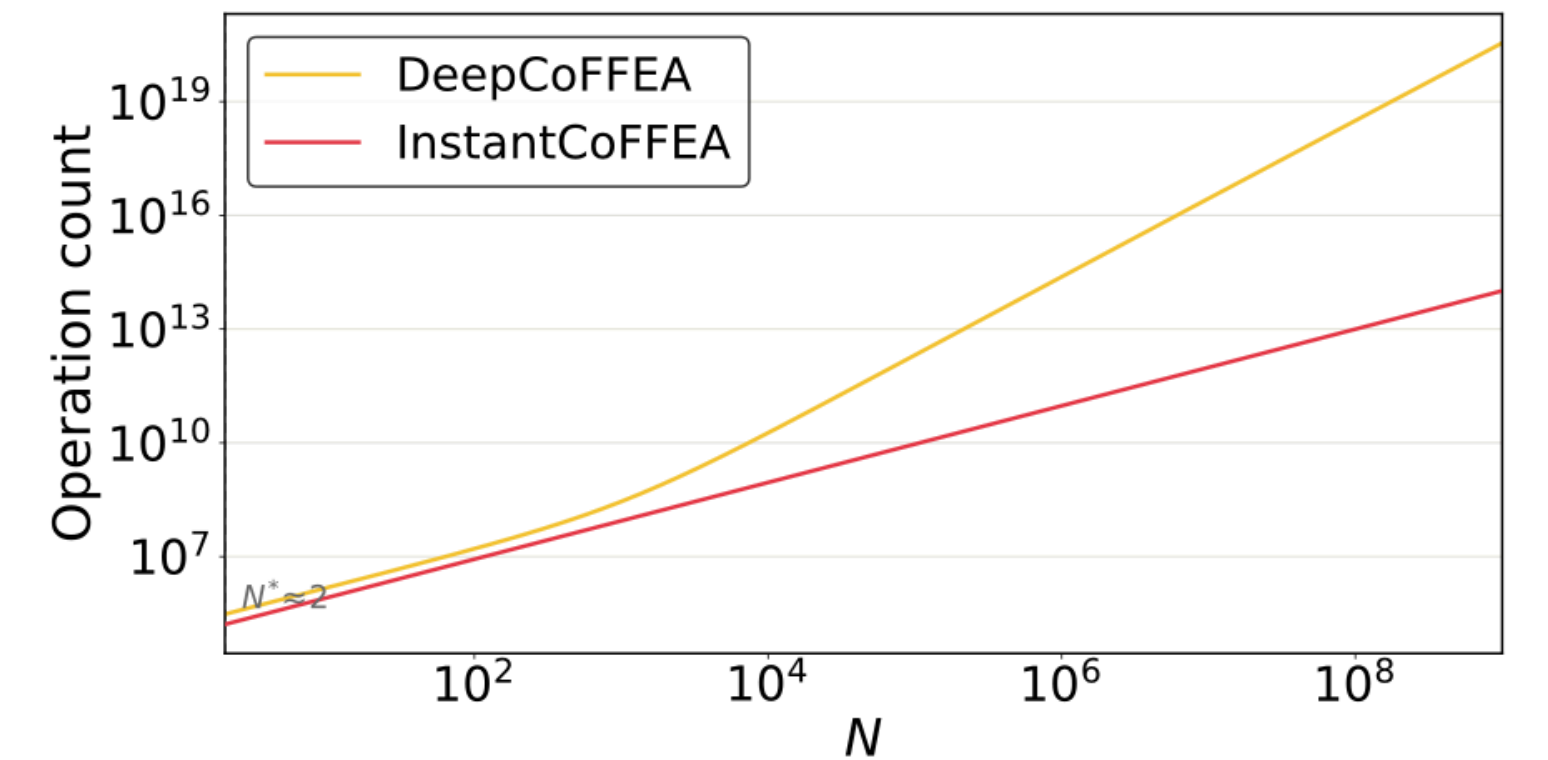


Fig. 10: Number of operations where $k = 1024, d = 16, p = 1, I_{\max} = 139$. Crossover point is $N = 2$.

Evaluation

Experiment setting:

- dataset: DeepCoFFEA (60k unique pairs) (for $N \geq 60k$, we use duplicated & jitter one)
- setting: lab setting & realistic setting

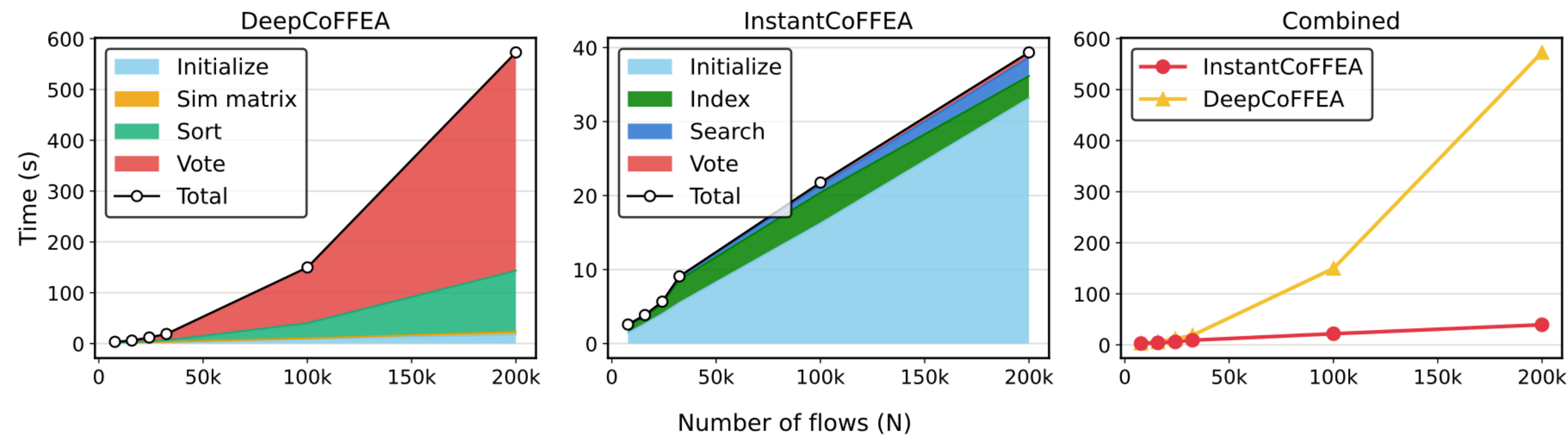
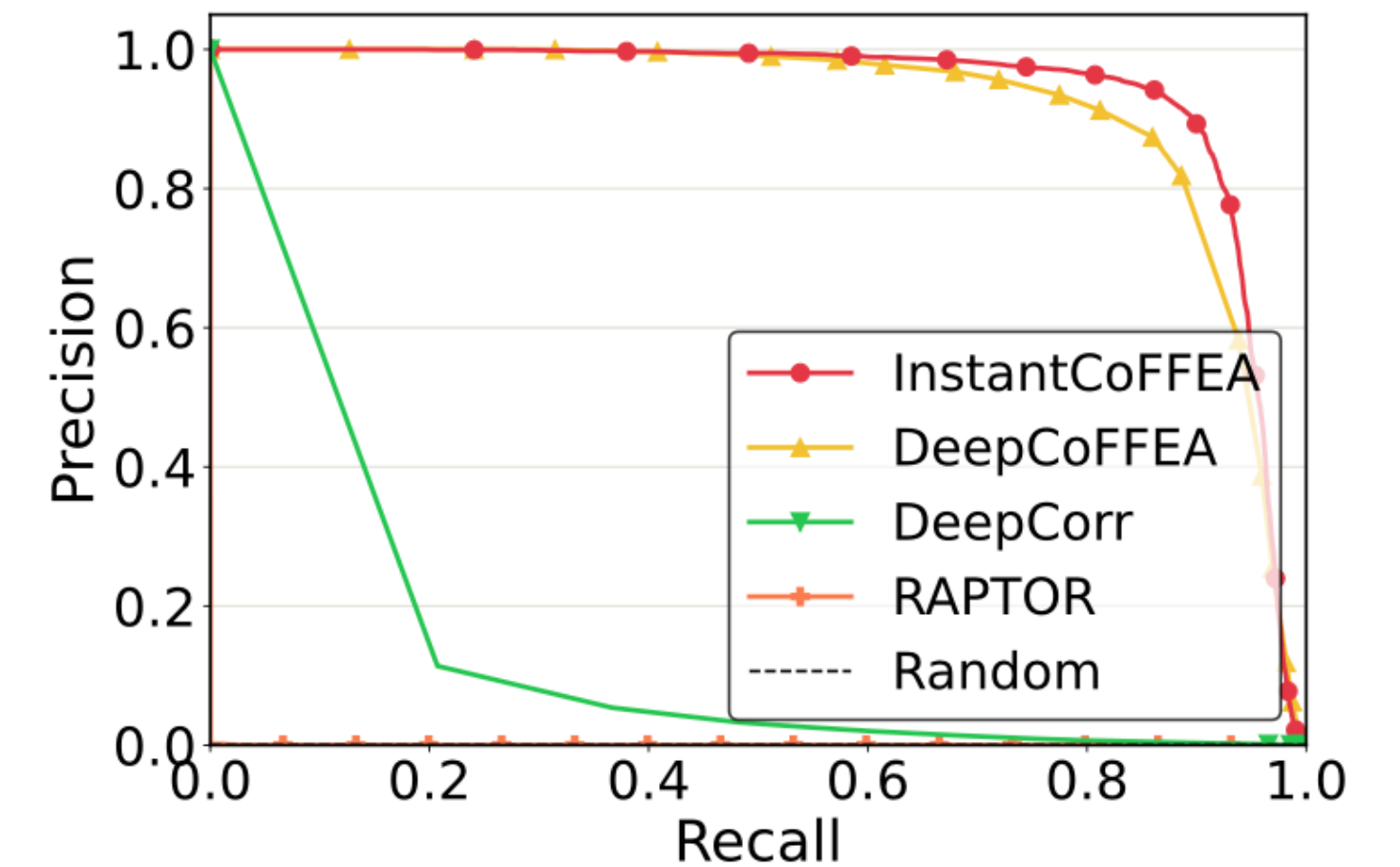


Fig. 4: Wall-clock time (seconds) breakdown of DeepCoFFEA (left) and InstantCoFFEA (middle) as N increases; the right panel compares their total runtimes.



(a) Lab setting

Evaluation

Ablation study for duplicated + jitter dataset

Table 5: InstantCoFFEA wall-clock time comparison between real and duplicated+jittered data by sample size N (Ratio = Jittered/Real)

N	Real (s)	Jittered (s)	Ratio
7,759	2.720 $\pm$ 0.771	2.640 $\pm$ 0.854	0.97x
15,866	5.877 $\pm$ 0.402	5.575 $\pm$ 0.402	0.95x
24,210	7.331 $\pm$ 0.702	6.707 $\pm$ 0.309	0.92x
62,727	14.885 $\pm$ 1.822	13.442 $\pm$ 0.839	0.90x

Ablation study for detection performance as N grows

Table 6: InstantCoFFEA and DeepCoFFEA’s PR-AUC as a function of pool size N

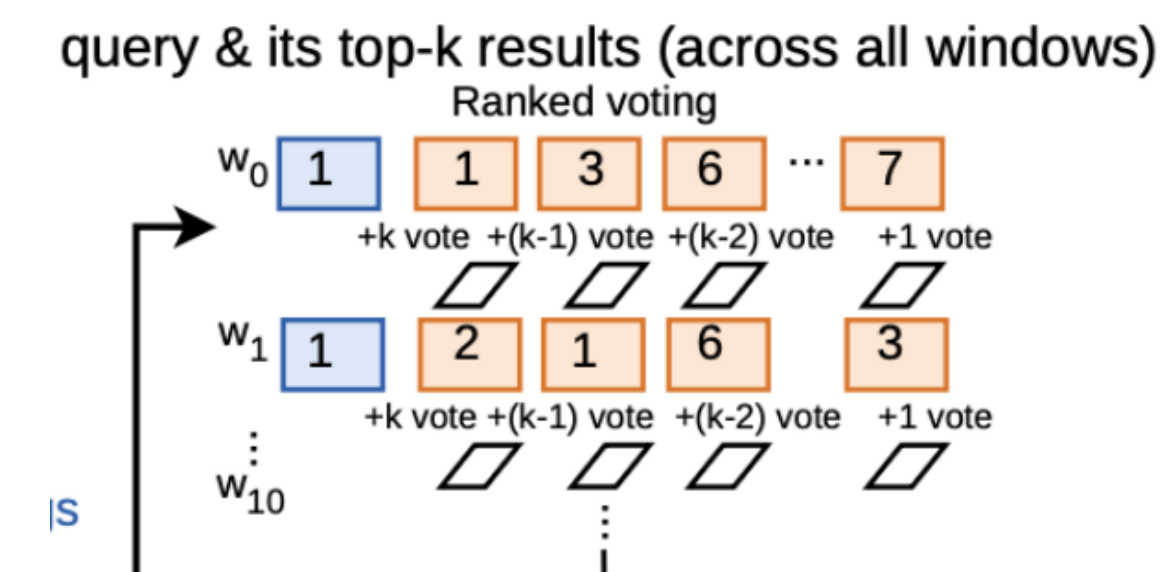
N	7,166	15,518	31,732	48,420	64,882
InstantCoFFEA	0.9315	0.9177	0.8837	0.8585	0.8397
DeepCoFFEA	0.8999	0.8910	0.8490	0.8234	0.7992

Evaluation

How InstantCoFFEA achieve better performance over DeepCoFFEA?

1. Each window stage, CAGRA's index minimize the loss

Setting	Method	Recall@ k								
		2	4	8	16	32	64	128	256	1024
Lab setting	ICF	0.3342	0.4102	0.4851	0.5623	0.6312	0.6952	0.7553	0.8254	0.8994
	DCF	0.3382	0.4148	0.4906	0.5693	0.6417	0.7103	0.7762	0.8316	0.9183



2. In window aggregation, ranked voting improve the performance

Lab setting (number of matching pairs=7804)								
FPR	7e-07	2e-06	4e-06	1e-05	3e-05	9e-05	4e-04	2e-03
DeepCoFFEA	3993	4803	5615	6336	6916	7323	7565	7705
InstantCoFFEA (binary voting)	2933	4469	5496	6261	6812	7175	7417	7680
InstantCoFFEA (ranked voting)	4452	5532	6340	6879	7169	7401	7578	7681

Simulation

2026-05-15일의 consensus document 를 이용해서, 매 시간에 1번, 총 24번, 10분 동안의 number of circuit in total network 를 기준으로, 각 adversary point 에서 얼마나 많은 flow 를 관찰하게 되는지 본다.

- Path selection tool: TorPS[1] 사용. (Ver 0.4.9 에 맞추어 몇가지 feature update)
- Number of circuit in total network: measurement paper [2,3] 에서 측정량 발체 [1.4M - 8.9M]
- 각 relay 정보: Tor consensus document 2026-05-15 의 정보 이용
- Relay 의 bandwidth 설정: 2,000MB/s, 4,000MB/s (각각 전체 대비 1.05%, 2.10%, TorPS 는 2.92% 사용했음)
- Network adversary 를 위한 AS, IXP, IXP organization 정보: CAIDA 에서 발체
- Destination 정보: 1. 전체에서 인기있는 사이트 Tranco list [4] / 2. 각 나라별로 주요 censored 된 사이트 이용 (Citizen Lab) [5]

[1] Johnson, A. Et al: Users get routed: traffic correlation on tor by realistic adversaries. In: Proceedings of the 2013 ACM SIGSAC conference on Computer & communications security. p. 337–348. CCS'13, Association for Computing Machinery, New York, NY, USA (2013). <https://doi.org/10.1145/2508859.2516651>

[2] Jansen, R., Johnson, A.: Safely measuring tor. In: Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security. CCS '16, Association for Computing Machinery, New York, NY, USA (2016), <https://doi.org/10.1145/2976749.2978310>

[3] Mani, A., Wilson-Brown, T., Jansen, R., Johnson, A., Sherr, M.: Understanding tor usage with privacy-preserving measurement. In: Proceedings of the Internet Measurement Conference 2018. p. 175–187. IMC '18, Association for Computing Machinery, New York, NY, USA (2018). <https://doi.org/10.1145/3278532>

[4] Le Pochat et al. Tranco: A research-oriented top sites ranking hardened against manipulation. In: Proceedings of the 26th Annual Network and Distributed System Security Symposium. NDSS 2019 (Feb 2019). <https://doi.org/10.14722/ndss.2019.23386>

[5] Citizen Lab: URL testing lists intended for discovering website censorship (2014), <https://github.com/citizenlab/test-lists>, <https://github.com/citizenlab/test-lists>

Simulation

Table 2: Simulated guard- and exit-side flow counts and compromise rates for relay- and network-level adversaries.

Adversary settings		1.4M	3M	6M	9M	Compromise rate
2,000 MB/s (relay)	Guard	13,889 $\pm$ 329	36,039 $\pm$ 843	60,014 $\pm$ 1,170	89,247 $\pm$ 1,731	0.0240% $\pm$ 0.0019%
	Exit	36,300 $\pm$ 1,480	93,117 $\pm$ 3,668	142,488 $\pm$ 4,637	213,833 $\pm$ 7,068	
4,000 MB/s (relay)	Guard	27,386 $\pm$ 647	70,246 $\pm$ 1,298	117,652 $\pm$ 2,156	177,226 $\pm$ 3,390	0.0903% $\pm$ 0.0058%
	Exit	70,438 $\pm$ 2,637	167,063 $\pm$ 5,387	278,506 $\pm$ 8,644	417,457 $\pm$ 12,948	
AS174 (AS)	Guard	255,968 $\pm$ 6,082	547,391 $\pm$ 13,052	1,094,497 $\pm$ 25,882	1,639,872 $\pm$ 38,554	2.9009% $\pm$ 0.0459%
	Exit	186,358 $\pm$ 3,756	399,120 $\pm$ 7,719	798,242 $\pm$ 15,837	1,195,155 $\pm$ 23,785	
Amsterdam IX (IXP)	Guard	282,661 $\pm$ 6,448	606,431 $\pm$ 13,743	1,210,416 $\pm$ 27,511	1,814,440 $\pm$ 40,992	12.9146% $\pm$ 0.0885%
	Exit	657,497 $\pm$ 19,472	1,409,261 $\pm$ 42,312	2,818,273 $\pm$ 84,131	4,222,067 $\pm$ 125,959	
Amsterdam IX B.V. (IXP Org)	Guard	282,726 $\pm$ 6,449	606,552 $\pm$ 13,747	1,210,688 $\pm$ 27,518	1,814,844 $\pm$ 41,002	12.9176% $\pm$ 0.0885%
	Exit	657,503 $\pm$ 19,472	1,409,273 $\pm$ 42,312	2,818,299 $\pm$ 84,131	4,222,105 $\pm$ 125,960	

Evaluation

Operating Time & Detection Performance

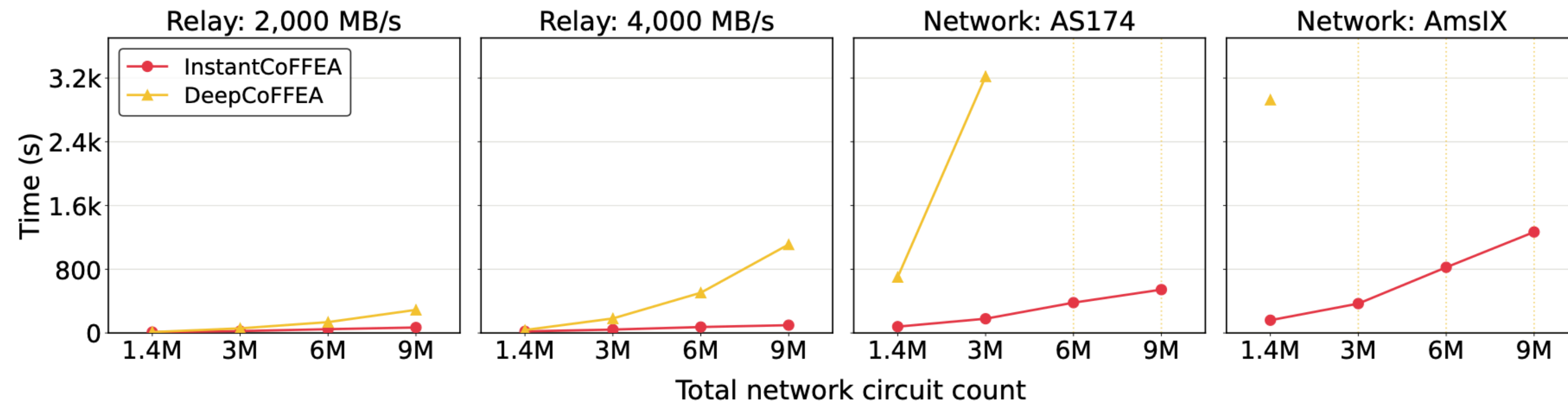
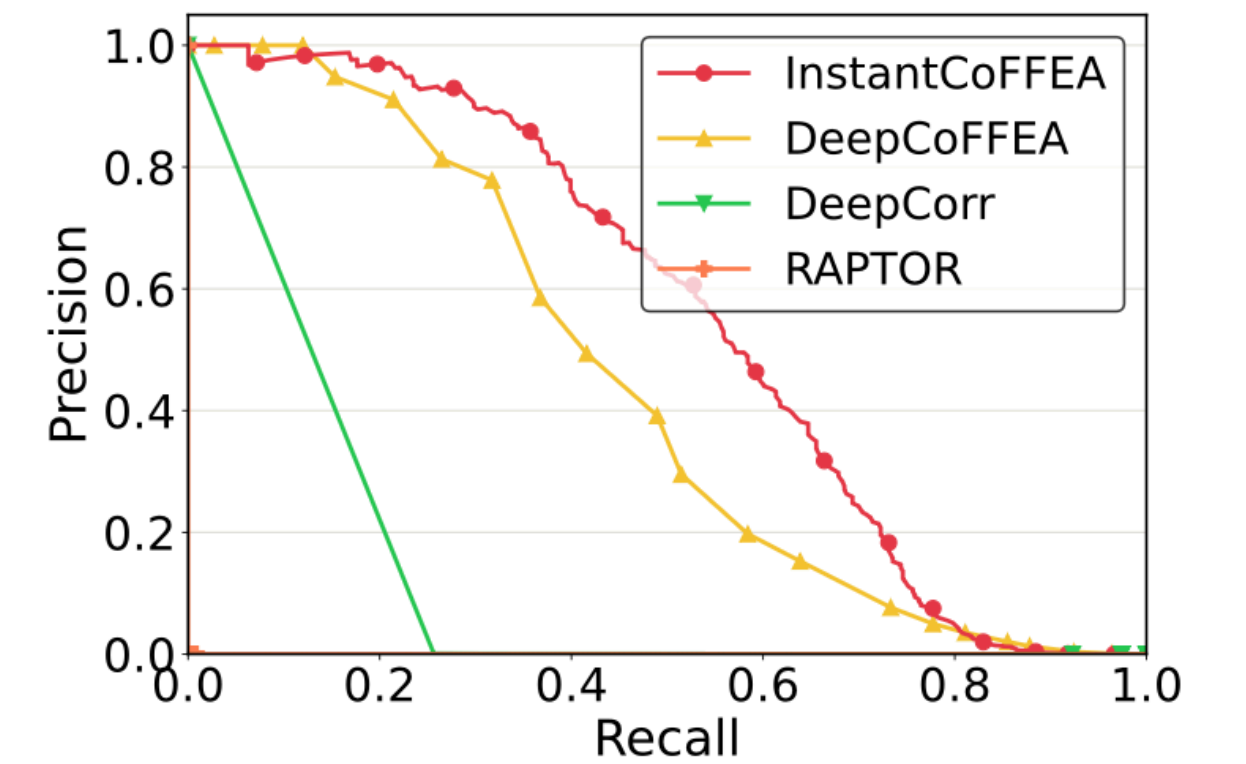


Fig. 6: Wall-clock time of InstantCoFFEA and DeepCoFFEA under realistic relay- and network-level adversary settings. Missing DeepCoFFEA points indicate out-of-memory (OOM) failures.



(b) Realistic setting

Conclusion

- We present InstantCoFFEA, which combines ANN search with ranked voting to improve both runtime and precision-recall over DeepCoFFEA.
- We also estimate realistic flow counts and base rates through simulation, showing that prior evaluations do not capture the sparsity of true matching pairs